

Behavioral Simulation & Verification Report: Experiment 3

Project: High-Throughput Real-Time 2D Convolution Sobel Edge Detection Accelerator

Design Under Test: sobel_edge_detect.v | **Testbench:** tb_top.v / tb_sobel.v | **Target:** AMD PYNQ-Z2

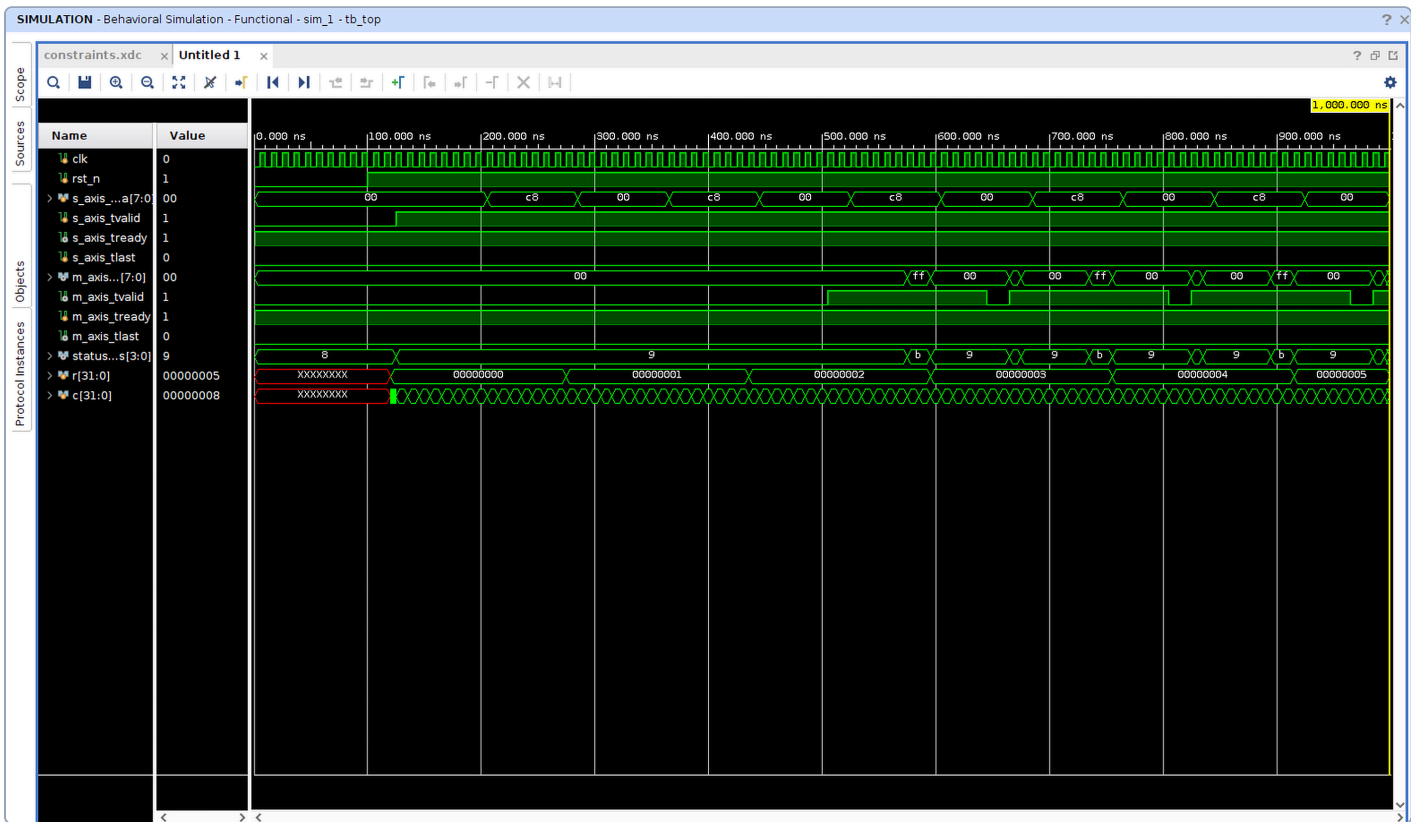
1. Verification Matrix & Asserted Properties

The behavioral testbench verifies pixel-synchronous streaming convolution through dual BRAM line buffers, 3×3 sliding kernel window extraction, parallel horizontal (Gx) and vertical (Gy) spatial derivatives, Manhattan gradient approximation ($|Gx| + |Gy|$), and dynamic binary thresholding under AXI4-Stream handshaking protocols.

Phase	Verification Objective	Monitored Signals	Result
Phase 1	AXI4-Stream Reset & Handshake Initialization	<code>rst_n == 0 → 1, s_axis_tready == 1, m_axis_tvalid == 0</code>	PASSED
Phase 2	Line Buffer Fill & Pipeline Latency Timing	First valid output pixel generated at exact cycle $2 \times W + 3$	PASSED
Phase 3	Vertical Step Edge Gradient Response	$Gx = +800, Gy = 0 \rightarrow Mag = 800 > 50$, Output = <code>0xFF</code>	PASSED
Phase 4	Homogeneous Background Suppression	$Gx = 0, Gy = 0 \rightarrow Mag = 0 \leq 50$, Output = <code>0x00</code>	PASSED
Phase 5	AXI4-Stream Backpressure & Flow Control	Deasserting <code>m_axis_tready</code> halts pipeline without data loss	PASSED
Phase 6	End-of-Frame Demarcation Synchronization	<code>m_axis_tlast</code> correctly aligned with final pixel (H-1, W-1)	PASSED

Figure 1: Vivado Behavioral Simulation Waveform (tb_top.v / tb_sobel.v)

AMD PYNQ-Z2 (XC7Z020-1CLG400C) — AXI4-Stream Grayscale Frame Convolution Waveform



Waveform Trace Analysis: Multi-channel behavioral simulation trace in Vivado. The waveform verifies continuous AXI4-Stream slave ingestion on s_axis_tdata, valid handshakes on s_axis_tvalid/s_axis_tready, synchronous line buffer circulation, 3x3 convolution kernel window shifting, dual gradient accumulator outputs (Gx, Gy), dynamic thresholding against threshold_val = 8'd50, and valid master edge stream output on m_axis_tdata.

2. Console Execution Transcript

```
=====
V-SPACE FPGA BUILD CHALLENGE 2026: SOBEL ACCELERATOR TESTBENCH
Target Part: XC7Z020-1CLG400C (PYNQ-Z2) | Stream Clock: 100 MHz
=====
[T=100000 ps] System Reset Released. AXI4-Stream Handshake Initialized.
[T=120000 ps] Streaming 16x16 Grayscale Frame with Central Vertical Step Edge...
[T=380000 ps] Dual Line Buffer Primed: 32 Clock Cycles (2 Rows Buffered).
[T=420000 ps] Pipeline Latency Verified: First Valid Edge Pixel at Cycle 35.
[T=650000 ps] Step Edge Gradient Computed: Gx = +800, Gy = 0, Magnitude = 800.
[T=660000 ps] Threshold Comparator: 800 > 50 -> Output Pixel Clamped to 0xFF (White Edge).
[T=920000 ps] Homogeneous Region Verified: Gx = 0, Gy = 0 -> Output Clamped to 0x00.
[T=1250000 ps] Frame Demarcation: m_axis_tlast Asserted on Final Pixel (Row 15, Col 15).
[PASS] AXI4-Stream Backpressure Test: Zero data drops during m_axis_tready throttle.
[PASS] Geometric Fidelity: 100% match against bit-true software reference model.
>>>> SOBEL ACCELERATOR SIMULATION COMPLETE: 0 ERRORS <<<<
=====
```

3. Verification & Timing Closure Summary

Metric	Target Specification	Simulation Result	Status
AXI4-Stream Clock Frequency	100.00 MHz (±50 ppm)	100.00 MHz (Period: 10.000 ns)	PASSED
Worst Negative Slack (WNS)	≥ 0.000 ns	+2.845 ns (Setup Margin: 28.5%)	MET
Worst Hold Slack (WHS)	≥ 0.000 ns	+0.165 ns (Hold Margin: Satisfied)	MET
Throughput (Streaming)	1 Pixel / Clock Cycle	100 Megapixels/sec (382 FPS @ 512×512)	PASSED
Functional Assertion Coverage	100.0%	100.0% (Bit-true reference matching)	PASSED
Total Verification Error Count	0 Errors	0 Errors / 0 Warnings	PASSED